

SPC-IRT II Draft Requirements

1. General Scope and Concept of Operations

1.1. **Purpose:** This document establishes the technical specifications for the Second-Generation Special Purpose Craft – Ice Rescue Transport (SPC-IRT II). The platform is engineered to standardize long-haul ice rescue airboat capabilities across the Great Lakes Area of Responsibility (AOR).

1.2. **Operational Roles:** The SPC-IRT II is engineered for a diverse mission set, prioritizing Search and Rescue (SAR), Living Marine Resources (LMR), and Other Law Enforcement (OLE) as its primary objectives. Beyond these core functions, the SPC-IRT serves in secondary capacities to transport crews through shallow water environments and operates as a versatile mobile command base, search vehicle, and temporary victim shelter. The SPC-IRT needs to be capable of operating during both day (light) and night (dark) to meet mission requirements.

1.3. **Operating Modes:** The SPC-IRT shall be able to operate in navigable waterways. The SPC-IRT shall smoothly transition between, and operate on dry land, solid clear/snow ice, broken pack/candle ice, mixed condition ice flows, and open water up to 10 NM offshore.

1.4. **Environmental Compliance:** The SPC-IRT shall comply with all applicable federal, state, local, and treaty environmental regulations, including air and noise emissions, fuel efficiency, bilge/oily water management, and the avoidance of harm to aquatic life. The contractor shall certify that the gasoline engine complies with applicable Environmental Protection Agency (EPA) exhaust emission regulations for marine spark-ignition engines.

2. Physical Characteristics & Environmental (Weather) Requirements

2.1. **Overall Length:** The overall length shall be a maximum of 25 feet (ft).

2.2. **Beam:** The beam shall be a maximum of 8 ft 6 in when trailered.

2.3. **Draft:** The draft shall be a maximum of 12 inches in fresh water in the Full Loaded Weight Condition.

2.4. **Speed:** The target speed shall be a minimum of 15 kts in seas up to 6 inches.

2.5. **Stopping Distance:** The SPC-IRT shall come to a complete stop from full speed in less than 100 feet on snow-covered ice (approximately 1/2 inch of snow on top of ice).

2.6. **Endurance / Range:** The SPC-IRT, on plane in open water (no ice), shall be capable of 5

hours of continuous operation with 10% fuel remaining when operating at the SPC-IRT's optimum planning speed for fuel efficiency.

2.7. Capacity: The SPC-IRT shall accommodate 6 Persons (full load condition is 3 crew, 3 passengers calculated at 220 lbs per person with PPE).

2.8. Temperature Range: The SPC-IRT shall operate in air temperatures ranging from -30°F to 115°F and water temperatures from freezing up to 80°F.

2.9. Wind & Sea State: The SPC-IRT shall operate in sustained true winds of 30 kts and shall survive gusts up to 40 kts. The SPC-IRT shall operate and survive in a sea state of up to 3 ft Significant Wave Height.

2.10. Visibility & Humidity: The SPC-IRT shall operate in 0% to 100% relative humidity, and in reduced visibility resulting from fog, snow, sleet or other heavy precipitation.

3. SPC-IRT Design, Functional, and Human Performance Requirements

3.1. Structural & Hull Design:

3.1.1. The hull form and structure shall demonstrate proven performance under the required operating conditions and the hull shall be constructed of a high-strength composite material specifically engineered to resist damage from ground contact and abrasive ice conditions.

3.1.2. The SPC-IRT shall possess a low center of gravity with distributed weight to prevent capsizing and minimize pressure on thin ice.

3.1.3. The SPC-IRT's systems and equipment design and installation shall comply with ABYC Standards and Technical Information Reports for Small Craft. The SPC-IRT shall possess positive buoyancy to meet basic flotation in accordance with ABYC H-8.

3.1.4. The SPC-IRT shall withstand the shock and vibration resulting from operation at 15 kts in 6-inch seas, and at minimum speed over rough ice including up to 2-foot vertical ice shelves.

3.1.5. The SPC-IRT shall incorporate structural hardpoints to support vessel-to-vessel towing operations (towing and being towed).

3.2. Mobility & Surface Transitions:

3.2.1. The SPC-IRT shall independently traverse over grass, dirt, snow, or ice for at least 1,000 yards without hull damage.

3.2.2. The SPC-IRT shall be capable of launching from and recovering to a trailer on soft water, hard water, and land, including frozen, and tundra environments. The SPC-IRT shall also be

capable of traversing unimproved embankments with gradients up to 15 degrees.

3.2.3. The SPC-IRT shall be capable of entering/departing water from land at angles up to 20 degrees (bow up), traversing 2-ft vertical ice compression ridges, and dropping from a 2-ft ice shelf into soft water in the Full Loaded Weight Condition without plunging (submerging) the bow.

3.2.4. The SPC-IRT shall maintain operational capability in up to 2 kts of current.

3.3. Cold Weather & Survivability:

3.3.1. The SPC-IRT shall be designed for a 10-year service life (when operated at 150 hours annually).

3.3.2. The SPC-IRT shall possess unassisted self-extrication capabilities if immobilized by ice and shall possess ice-anchoring capabilities (ice anchor and winch).

3.3.3. The SPC-IRT shall have a block heater powered by the trailer/transport vehicle, cabin heating for the crew, windshield de-icing/defrosting systems, and a means to clear windshields of water build-up from mist, rain, and sea-spray.

3.3.4. Propeller/fan and cooling systems shall feature guarding or cages to prevent foreign object damage (FOD) and provide protection for personnel safety.

3.3.5. The SPC-IRT shall feature underway dewatering access for all bilges and voids, active bilge ventilation for flammable vapor mitigation, and at least two illuminated emergency egress pathways.

3.4. Ergonomics & Human Factors:

3.4.1. The SPC-IRT shall accommodate a 5th percentile female up to a 95th percentile male personnel (ASTM F1166) in full cold weather PPE.

3.4.2. The SPC-IRT shall feature a semi-enclosed cabin structure with canvas/fabric enclosures to protect the crew from weather. The design shall maximize external visibility.

3.4.3. The SPC-IRT shall have six seats total with crew restraints. The two forward-facing shock-mitigating seats (Coxswain and Navigator) shall feature swivel capability and shall actively mitigate shock for personnel ranging from the 5th percentile female to the 95th percentile male in full cold weather PPE.

3.4.4. The SPC-IRT design shall ensure no more than two crew members are needed to recover an incapacitated person from the water.

3.4.5. Handholds shall be dedicated and strategically placed to prevent knee/leg impacts while underway. Hatch and handrail design shall facilitate safe movement. All controls shall be usable by personnel wearing heavy PPE.

3.4.6. The air rudders shall feature the US Coast Guard stripe and emblem (blue, white, and orange stripes).

4. Electronics, Communications, and Navigation Requirements

4.1. Internal Communications:

4.1.1. The SPC-IRT shall feature a single-circuit comms system at each of the four crew stations.

4.1.2. The system shall conform to FCC regulations and shall not interfere with any other internal or external communications control system.

4.1.3. The system shall allow users to switch from a wired connection to a wireless system featuring VOX (Voice-Activated Transmit).

4.1.4. The system shall provide full-duplex comms, with noise reduction capability, and wearable transceivers allowing full range of motion.

4.1.5. Transceiver batteries shall be resistant to memory effects and provide a minimum of 8 hours of operation with a 5-5-90 duty cycle (5% Transmit, 5% Receive, 90% Standby). An additional spare battery shall be provided for each crewman transceiver.

4.1.6. Communications controls shall be easily accessible to the coxswain. The internal communications system shall be integrated with the VHF DSC radio (GX-6000).

4.2. External Communications:

4.2.1. The SPC-IRT shall be equipped with a Standard Horizon GX-6000 VHF Class D Marine Band radio with Digital Selective Calling (DSC) capability. The Standard Horizon GX-6000 shall be integrated with the SINS-2 navigation system to support DSC functionality.

4.2.2. The SPC-IRT shall have a forward-facing exterior HAIL/PA horn/speaker with listen-back capability, integrated with the Standard Horizon GX-6000 and installed per OEM requirements.

4.2.3. All installed electrical/electronic equipment shall be capable of withstanding temporary fresh water submersion, enclosed in watertight enclosures, or otherwise protected from moisture associated with spray, wave splashing, condensation, or other sources.

4.3. Navigation System (USCG SINS-2):

The SPC-IRT shall be equipped with the following integrated USCG SINS-2 equipment.

Navigation controls and displays shall be easily accessible and viewable by the coxswain. The contractor shall provide electronic charts for testing at their facilities.

4.3.1. One (1) Raymarine Axiom 2 Pro 12 RVM 12" inch Multifunctional Display (P/N E70656).

4.3.2. One (1) Raymarine Axiom 2 Pro 9 RVM 9" inch Multifunctional Display (P/N E70654).

4.3.3. One (1) Raymarine EV-1 Heading Sensor (P/N E70096).

4.3.4. One (1) Raymarine RS-150 Global Navigation Satellite System (GNSS) Receiver (P/N E70310).

4.3.5. One (1) Actisense NGW-1 National Marine Electronics Association (NMEA) 0183 to 2000 Gateway (Raymarine P/N A80455).

4.3.6. One (1) Radome 4kW 24" HD Radar (P/N E92143).

4.3.7. One (1) RNS-5 Raynet Network Switch (P/N A80731).

4.4. Automatic Identification System (USCG AIS-2):

The SPC-IRT shall be equipped with an AIS suite that is integrated with SINS-2. The AIS system shall include:

4.4.1. One (1) Raymarine AIS5000 Transponder Bundle (Transponder, GPS antenna, Power/Control/RF Coax cables, Mounting Bracket, Junction Box, and Manual) (Raymarine P/N E70529).

4.4.2. One (1) VHF Antenna (Raymarine P/N A80590).

4.4.3. One (1) DeviceNet to STng Cable (Raymarine P/N A06075).

4.5. Sensors & Lighting:

4.5.1. The SPC-IRT shall feature an Infrared (IR) Camera for daytime and nighttime use with pan/tilt capability, gyro stabilization, high-definition display, thermal/IR and visible light modes, and optical zoom, integrated into the SINS-2 display.

4.5.2. Exterior lighting shall include forward floodlights, 360-degree side floodlights, mounted blue law enforcement lights, and handheld spotlights with a minimum 110,000 Candela.

4.5.3. Interior lighting shall include dimmable to zero cabin lights that can be switched from Off to White or Red. They shall be dimmable to zero in both the white and red positions.

4.5.4. Navigation lights and sound signals shall comply with 72 COLREGS and Inland Navigation Rules Annex III, and shall require minimal service in icing conditions.

4.5.5. The SPC-IRT shall have audible and visual engine alarm systems.

5. Trailer and Outfitting Requirements

5.1. Trailer:

5.1.1. The trailer shall feature a tilting design equipped with an integrated electric winch (rated to 150% of the vessel's weight) that is lanyard/remote operated.

5.1.2. The trailer shall be rated for standard use on US highways at 65 mph. The entire loaded assembly shall not exceed 10,000 lbs.

5.2. Standard Outfit:

5.2.1. The SPC-IRT shall provide the following equipment with designated onboard storage locations:

5.2.1.1. Expandable 8-foot Boat Hook

5.2.1.2. Binoculars (7x50)

5.2.1.3. Two Dry Chem. Fire Extinguishers (Per ABYC A-4) installed on brackets (fore/aft)

5.2.1.4. Paddle

5.2.1.5. Mooring Lines: (2) 15-ft ½ inch double braided nylon (DBN), (2) 25- ft ½ inch DBN

5.2.1.6. Fabric briefcase bag for CG supplied Navigation Kit

5.2.1.7. Water Rescue Throw Bag w/ 75 feet of 7/16" high visibility floating line

5.2.1.8. Life Ring w/ mounting bracket, 24", orange

5.2.1.9. Floating marker light

5.2.1.10. Handheld rechargeable searchlight, (minimum 110,000 candela)

5.2.1.11. Personal Flotation Devices (2 x USCG Type 1, Adult)

5.2.1.12. Tool Kit contained in a watertight, airtight, dustproof, chemical resistant, corrosion proof case, approx. 14 inches x 12 inches x 6 inches, with inserts and shall at a minimum have the following tools:

5.2.1.12.1. Metric Combination Wrench Set: 10mm, 11mm, 12mm, 13mm, 14mm, 15mm, 16mm, and 17mm in a durable case;

5.2.1.12.2. SAE Combination Wrench Set: 1/4, 5/16, 3/8, 7/16, 1/2, 9/16, 5/8, and 11/16 in. in a durable case;

5.2.1.12.3. 3/8 in drive SAE 6 or 12-point Socket Set: 1/4, 5/16, 3/8, 7/16, 1/2, 9/16, 5/8, 11/16, and 3/4 in. in a durable hard plastic case;

5.2.1.12.4. 3/8 in. drive Metric 6 or 12-point Socket Set 10mm, 12mm, 13mm, 14mm, 15mm, 16mm, 17mm, 18mm, and 19mm in a durable hard plastic case;

- 5.2.1.12.5. 3/8 in. drive ratchet with 3 in. and 6 in. extension bar in a durable hard plastic case;
- 5.2.1.12.6. 10 in. Adjustable Wrench;
- 5.2.1.12.7. Strap Wrench Set (two Piece);
- 5.2.1.12.8. 10 in. Curved Jaw Locking Pliers;
- 5.2.1.12.9. 10 in. Straight Jaw Pushlock Pliers;
- 5.2.1.12.10. 10 in. Straight Jaw Pliers;
- 5.2.1.12.11. 10-in-1 Ratcheting Multi-bit Screwdriver Set;
- 5.2.1.12.12. 290 lumen LED Rechargeable Waterproof Spot-Beam Flashlight;
- 5.2.1.12.13. 1/8 inch x 1/8 inch x 7.5 inch black poly cable ties, 50 Pack;
- 5.2.1.12.14. One roll (20 yards) of 2-inch wide duct tape; and
- 5.2.1.12.15. Electrical Tape (20 yard, roll)

6. Logistics, Maintenance, and Training Requirements

6.1. Documentation:

- 6.1.1. The contractor shall deliver a Boat Information Book (BIB) per ABYC T-24.
- 6.1.2. The contractor shall deliver full technical drawings (including hull lines, flotation calculations, structural arrangements, doors/hatches, flotation materials, propulsion, fuel, electrical one-line diagrams, battery details, lighting, electronics/sensors, bilge, stowage, canvas, and trailer).
- 6.1.3. The contractor shall deliver a Master Equipment List, Recommended Spare Parts List, and a List of Special Tools and Long Lead Parts.

6.2. Maintainability & Supportability:

- 6.2.1. The SPC-IRT shall be configured to support efficient maintenance and repair and be provided with predicted maintenance needs. In selecting equipment reliable and low-maintenance shall be a consideration with adequate physical and visual access for maintenance considered in establishing the configuration.
- 6.2.2. The contractor shall make all parts and components on the SPC-IRT available for purchase.
- 6.2.3. The contractor shall develop and deliver a 150-Hour Maintenance Requirements List (MRL). The contractor shall provide all maintenance parts, consumables, and special tools necessary to perform the 150-Hour MRL tasks for each boat.

6.2.4. The contractor shall provide a list of all special tools and test equipment, detailing the description, the system it supports, and the source of supply. A special tool shall be defined as any tool not commercially available or requiring specific fabrication.

6.2.5. The contractor shall provide one copy of any engine diagnostic software licenses, unique diagnostic dongles, and fabrication rights/specs for special tools per boat.

6.3. Training:

6.3.1. Familiarization (FAM) Training covering safe boat handling and operation, emergency casualty control, and basic operational procedures shall be provided at each delivery location within 5 days of delivery.

Appendix A: Definitions

A.1. **Sea State:** All “Sea State” references shall apply to the Bretschneider Wave Analysis and Forecasting Nomogram.

A.2. **Lightship Weight Condition:** The weight of the SPC-IRT and outfit items without fuel or people. Engine/propulsion lubricants shall be at normal operating levels.

A.3. **Performance Weight Condition:** Lightship Weight Condition plus full fuel plus a 3-person crew. Crew and passenger weights shall be calculated at 220 pounds per person.

A.4. **Full Load Weight Condition:** Performance Weight Condition plus 3 passengers. Crew and passenger weights shall be calculated at 220 pounds per person.

A.5. **Operate:** Operating shall include wave height conditions below survivability limits. It shall involve the application of various SPC-IRT speeds that ensure bow immersions do not cause flooding which leads to reduced stability. It shall also include the continuous ability to maintain motions and accelerations at levels safe for the crew, machinery, systems, hull structure, and watertight integrity.

A.6. **Survive:** Survive shall be defined as the SPC-IRT’s ability to maintain upright stability, flotation, and structural and watertight integrity. It shall include the crew’s ability to operate and maneuver the SPC-IRT to return to a safe haven. In the event no safe haven is available, or navigation is not possible, the SPC-IRT shall provide a safe haven for the crew.

A.7. **Personnel:** All references to personnel shall include the 5th percentile female through the 95th percentile male in cold weather gear.